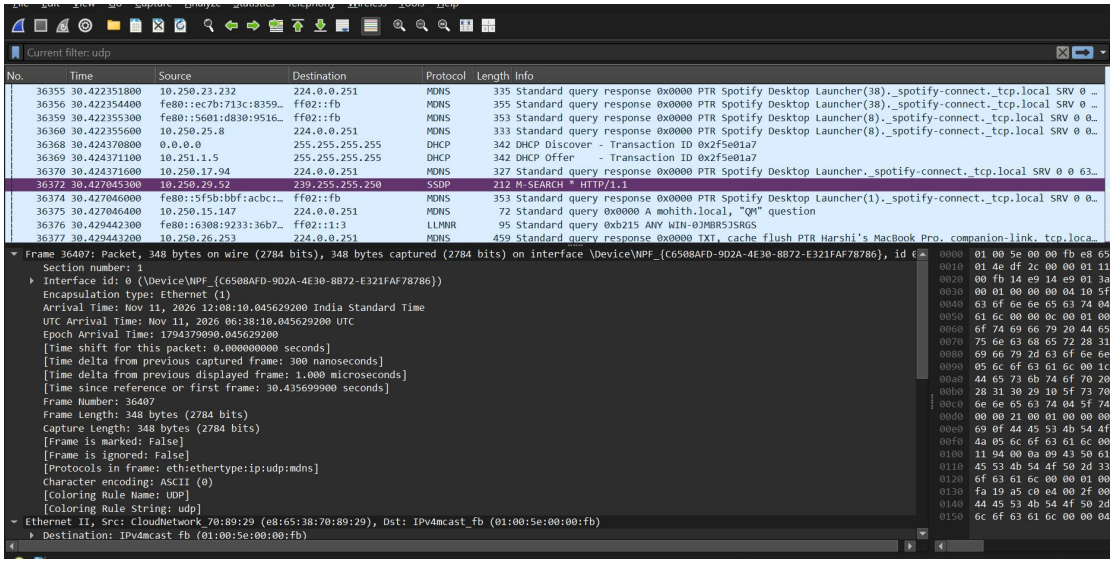
SKILL - 01

Step 1: Install Wireshark

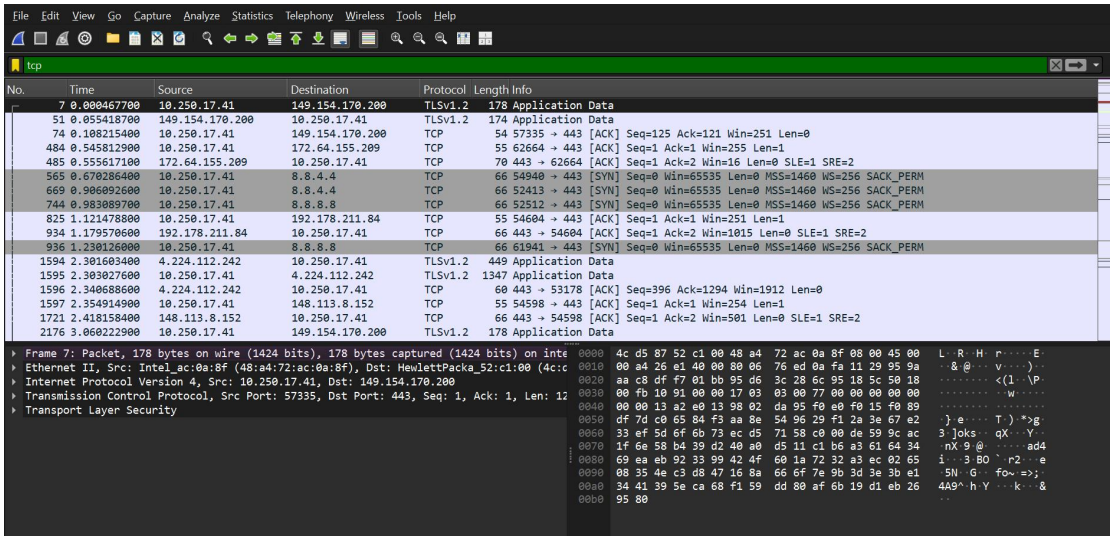
1. Open the browser.
2. Install Wireshark Wireshark • Go Deep | Download
3. Install it using the default settings.
4. Open Wireshark

Step 2: Generate network traffic
While Wireshark is capturing:
1. Open a new browser tab.
2. Visit a website such as https://www.google.com.
3. Open two or three different websites.
4. Wait for approximately 30 seconds.
5. Return to Wireshark.
6. Click the red Stop button.

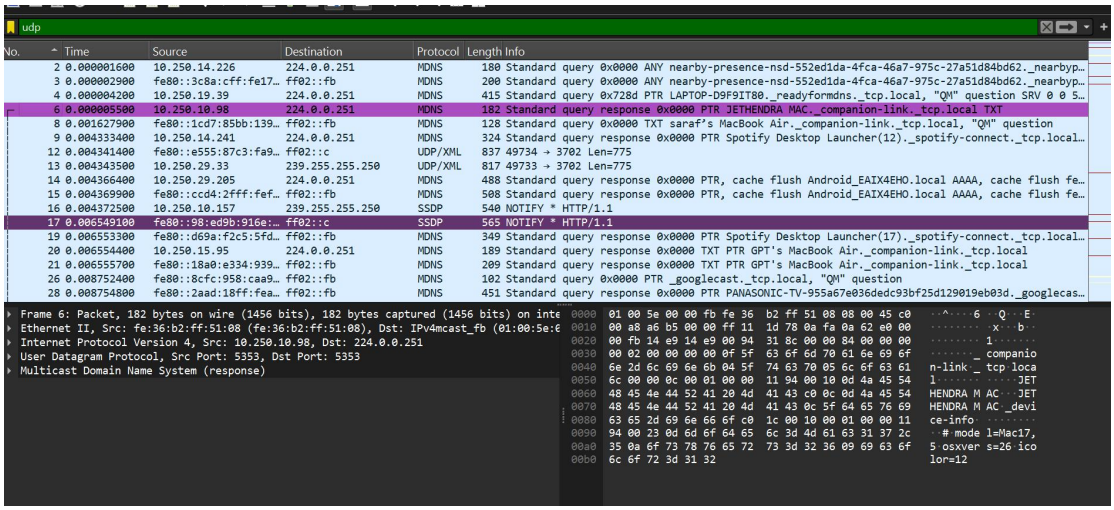


Step 3: Apply display filters
In the Wireshark filter bar, enter each filter separately and press Enter

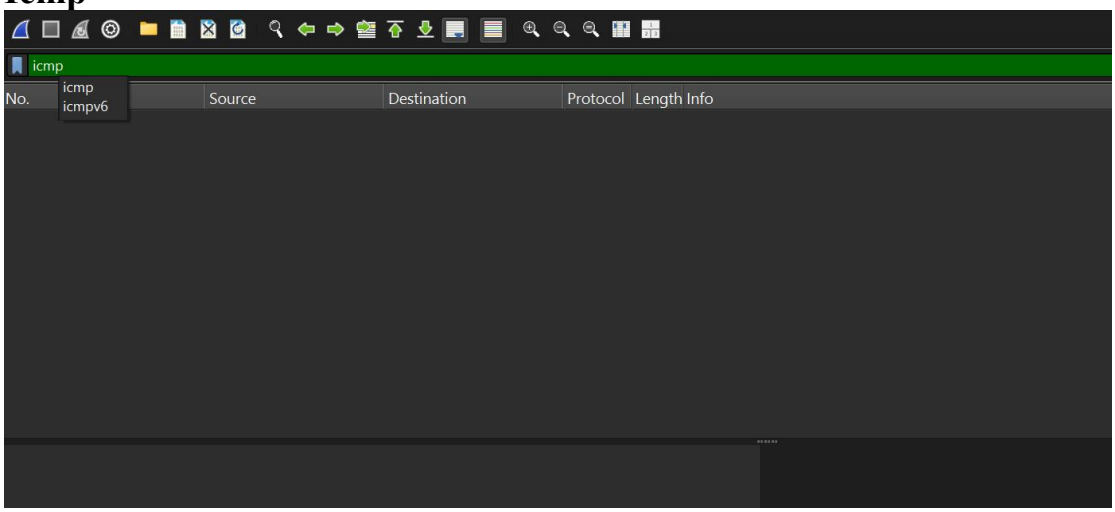
TCP



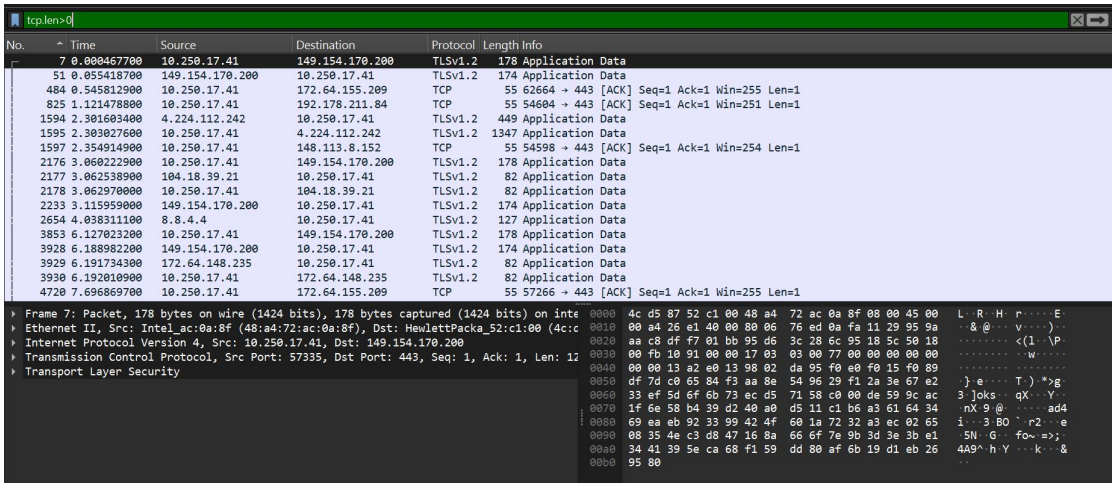
UDP



Icmp

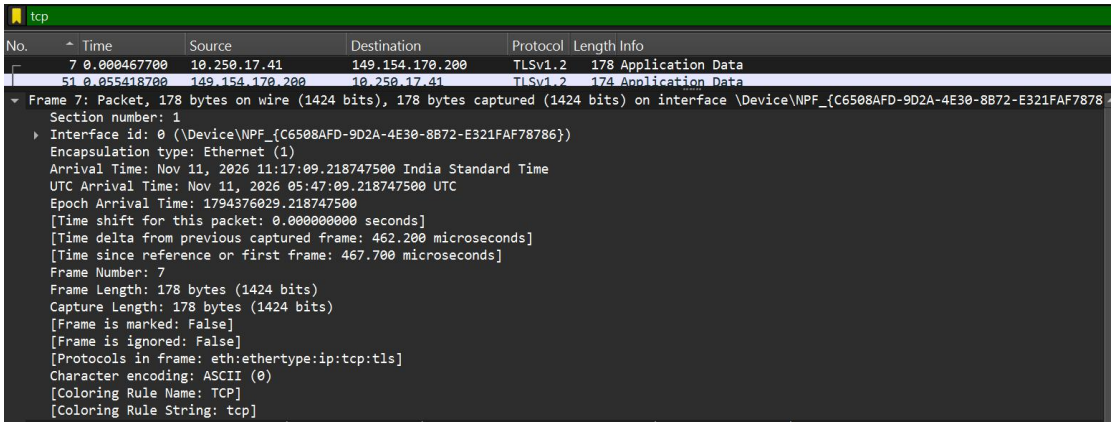


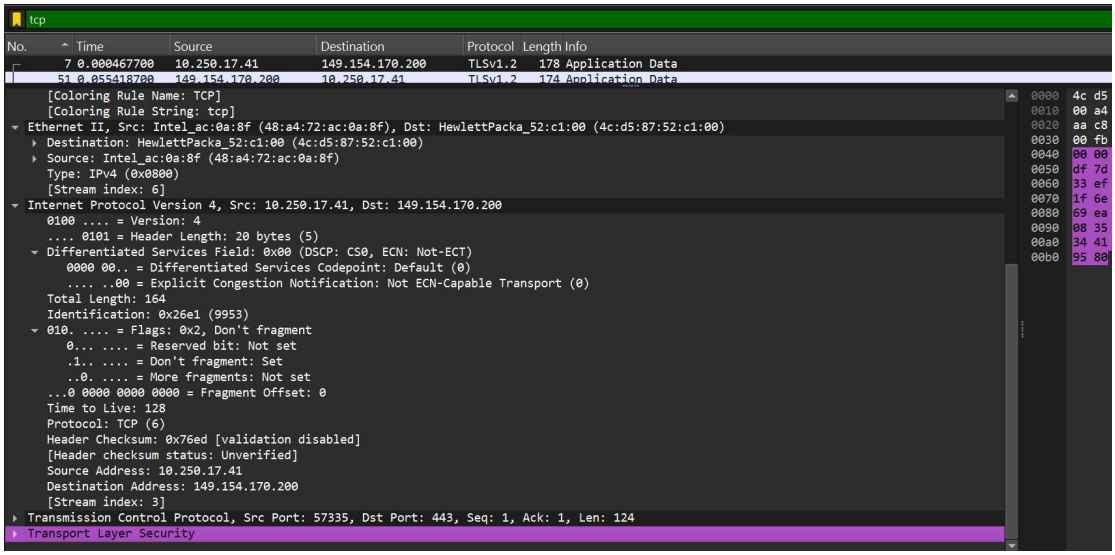
Tcp.len>0



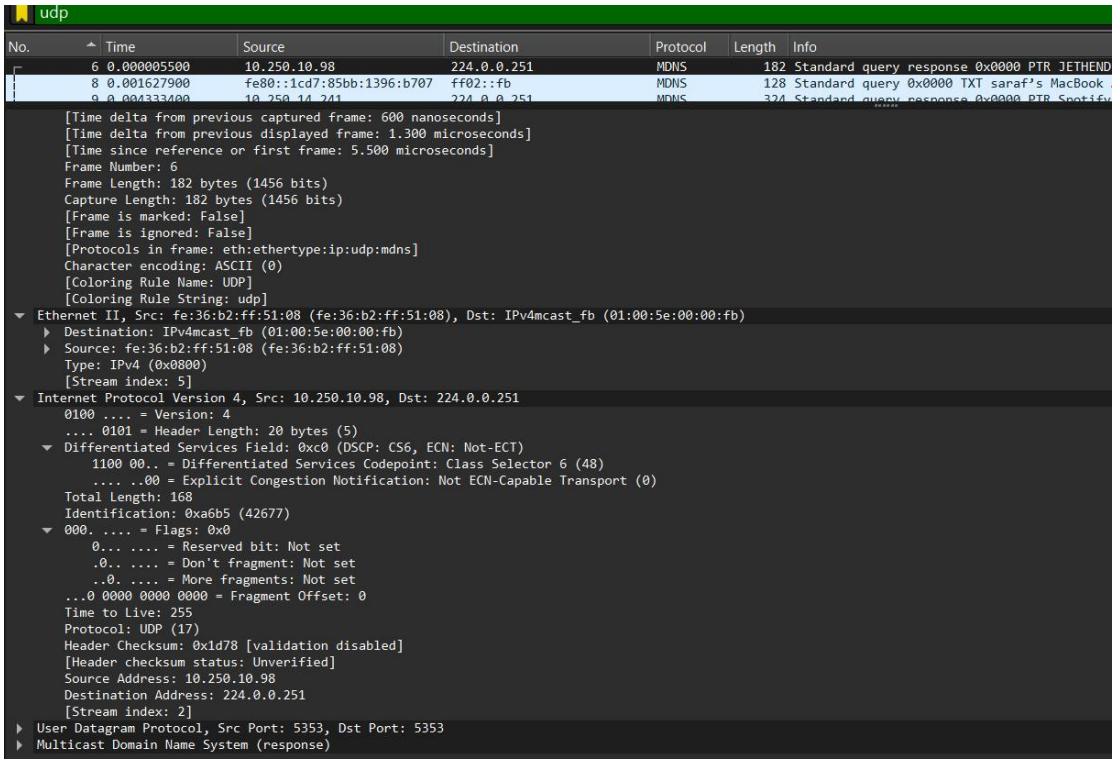
Step 4: Inspect packet headers
Select any TCP packet from the packet list.
In the packet details pane, expand these sections:
1. Frame
2. Ethernet II
3. Internet Protocol Version 4
4. Transmission Control Protocol

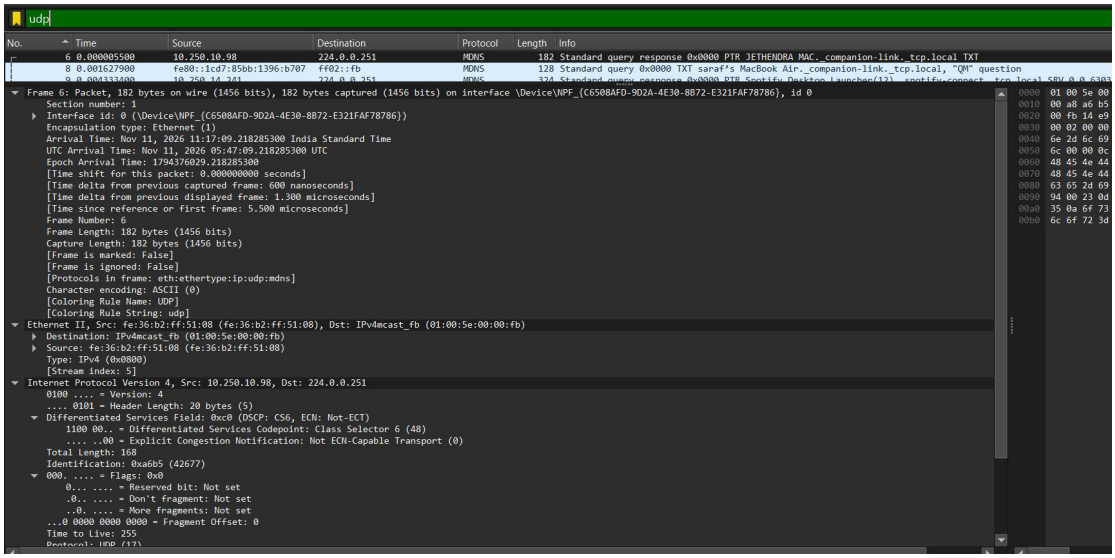
TCP




tcp
No. Time Source Destination Protocol Length Info
7 0.000467700 10.250.17.41 149.154.170.200 TLSv1.2 178 Application Data
51 0.055418700 149.154.170.200 10.250.17.41 TLSv1.2 174 Application Data
[Coloring Rule Name: TCP]
[Coloring Rule String: tcp]
Ethernet II, Src: Intel_ac:0a:8f (48:a4:72:ac:0a:8f), Dst: HewlettPacka_52:c1:00 (4c:d5:87:52:c1:00)
Destination: HewlettPacka_52:c1:00 (4c:d5:87:52:c1:00)
Source: Intel_ac:0a:8f (48:a4:72:ac:0a:8f)
Type: IPv4 (0x0800)
[Stream index: 6]
Internet Protocol Version 4, Src: 10.250.17.41, Dst: 149.154.170.200
0100 .... = Version: 4
.... 0101 = Header Length: 20 bytes (5)
Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
0000 00.. = Differentiated Services Codepoint: Default (0)
.... ..00 = Explicit Congestion Notification: Not ECN-Capable Transport (0)
Total Length: 164
Identification: 0x26e1 (9953)
010. .... = Flags: 0x2, Don't fragment
0... .... = Reserved bit: Not set
.1.. .... = Don't fragment: Set
..0. .... = More fragments: Not set
...0 0000 0000 0000 = Fragment Offset: 0
Time to Live: 128
Protocol: TCP (6)
Header Checksum: 0x76ed [validation disabled]
[Header checksum status: Unverified]
Source Address: 10.250.17.41
Destination Address: 149.154.170.200
[Stream index: 3]
Transmission Control Protocol, Src Port: 57335, Dst Port: 443, Seq: 1, Ack: 1, Len: 124
Transport Layer Security
0000 4c d5
0010 00 a4
0020 aa c8
0030 00 fb
0040 00 00
0050 df 7d
0060 33 ef
0070 1f 6e
0080 69 ea
0090 08 35
00a0 34 41
00b0 95 80

UDP


udp
No. Time Source Destination Protocol Length Info
6 0.000005500 10.250.10.98 224.0.0.251 MDNS 182 Standard query response 0x0000 PTR JETHEND
8 0.001627900 fe80::1cd7:85bb:1396:b707 ff02::fb MDNS 128 Standard query 0x0000 TXT saraf's MacBook
[Time delta from previous captured frame: 600 nanoseconds]
[Time delta from previous displayed frame: 1.300 microseconds]
[Time since reference or first frame: 5.500 microseconds]
Frame Number: 6
Frame Length: 182 bytes (1456 bits)
Capture Length: 182 bytes (1456 bits)
[Frame is marked: False]
[Frame is ignored: False]
[Protocols in frame: eth:ethertype:ip:udp:mdns]
Character encoding: ASCII (0)
[Coloring Rule Name: UDP]
[Coloring Rule String: udp]
Ethernet II, Src: fe:36:b2:ff:51:08 (fe:36:b2:ff:51:08), Dst: IPv4mcast_fb (01:00:5e:00:00:fb)
Destination: IPv4mcast_fb (01:00:5e:00:00:fb)
Source: fe:36:b2:ff:51:08 (fe:36:b2:ff:51:08)
Type: IPv4 (0x0800)
[Stream index: 5]
Internet Protocol Version 4, Src: 10.250.10.98, Dst: 224.0.0.251
0100 .... = Version: 4
.... 0101 = Header Length: 20 bytes (5)
Differentiated Services Field: 0xc0 (DSCP: CS6, ECN: Not-ECT)
1100 00.. = Differentiated Services Codepoint: Class Selector 6 (48)
.... ..00 = Explicit Congestion Notification: Not ECN-Capable Transport (0)
Total Length: 168
Identification: 0xa6b5 (42677)
000. .... = Flags: 0x0
0... .... = Reserved bit: Not set
.0.. .... = Don't fragment: Not set
..0. .... = More fragments: Not set
...0 0000 0000 0000 = Fragment Offset: 0
Time to Live: 255
Protocol: UDP (17)
Header Checksum: 0x1d78 [validation disabled]
[Header checksum status: Unverified]
Source Address: 10.250.10.98
Destination Address: 224.0.0.251
[Stream index: 2]
User Datagram Protocol, Src Port: 5353, Dst Port: 5353
Multicast Domain Name System (response)



Record the following values:

Parameter	TCP Packet	UDP Packet
Frame Length	178 bytes	182 bytes
Ethernet Header Length	14 bytes	14 bytes
Source MAC Address	48:a4:72:ac:0a:8f	fe:36:b2:ff:51:08
Destination MAC Address	4c:d5:87:52:c1:00	01:00:5e:00:00:fb
IPv4 Header Length	20 bytes	20 bytes
IPv4 Total Length	164 bytes	168 bytes
Transport Protocol	TCP	UDP
Transport Header Length	20 bytes	8 bytes
TCP Segment / UDP Datagram Length	144 bytes	148 bytes
Payload Length	124 bytes	140 bytes

Step 5: Calculate Header Overhead

1. TCP Packet

Parameter	Observed value
Frame Length	**178 bytes**
Ethernet Header	**14 bytes**
IPv4 Header	**20 bytes**
TCP Header	**20 bytes**
TCP Data Length	**124 bytes**
IP Total Length	**164 bytes**

Calculation:

Total header overhead:

$$14 + 20 + 20 = 54 \text{ bytes}$$

Total frame size:

$$54 + 124 = 178 \text{ bytes}$$

Payload efficiency:

$$124 / 178 * 100 = 69.66\%$$

2. UDP Packet

Parameter	Observed value
Frame Length	**182 bytes**
Ethernet Header	**14 bytes**
IPv4 Header	**20 bytes**
UDP Header	**8 bytes**
UDP Data Length	**140 bytes**
IP Total Length	**168 bytes**

Calculation:

Total header overhead:

14 + 20 + 8 = 42 bytes

Total frame size:

42 + 140 = 182 bytes

Payload efficiency:

140 / 182 * 100 = 76.92

Final results

Parameter	TCP	UDP
Frame Length	**178 B**	**182 B**
Header Overhead	**54 B**	**42 B**
Payload	**124 B**	**140 B**
Payload Efficiency	**69.66%**	**76.92%**

Step 6: Fill in your observation table

Parameter	TCP Packet	UDP Packet
Frame Length	**178 bytes**	**182 bytes**
Ethernet Header Length	**14 bytes**	**14 bytes**
IP Header Length	**20 bytes**	**20 bytes**
Transport Header Length	**20 bytes**	**8 bytes**
IP Total Length	**164 bytes**	**168 bytes**
Payload Length	**124 bytes**	**140 bytes**